

CFD 2025-26 – Lab 10 – solution

Exercise 1

The implementation of the full solver is provided in `CFDlab10.ipynb`.

- 1 Denoting by $\mathbf{u}_{\text{in}}$ the inlet velocity and by $\Gamma_w = \partial\Omega \setminus (\Gamma_{\text{in}} \cup \Gamma_{\text{out}})$ the lateral walls, we define the following spaces:

$$V = \{\mathbf{v} \in [H^1(\Omega)]^2 : \mathbf{v} = \mathbf{u}_{\text{in}} \text{ on } \Gamma_{\text{in}}, \mathbf{v} = \mathbf{0} \text{ on } \Gamma_w\},$$

$$V_0 = [H_{\partial\Omega \setminus \Gamma_{\text{out}}}^1(\Omega)]^2, \quad Q = H_{\Gamma_{\text{out}}}^1(\Omega),$$

where we remark that the pressure space Q is a subspace of $\underline{H^1(\Omega)}$: this is required for step 2 of the Chorin-Temam scheme to be well-posed.

Introducing proper FE spaces $V_h \subset V$, $V_{0,h} \subset V_0$, $Q_h \subset Q$, and a discrete lifting $\mathcal{R}_h \mathbf{u}_{\text{in}} \in V_h$ of the inflow data, and applying the FEM, we can obtain the fully discrete numerical method:¹

Given $\mathbf{u}_h^0 = \mathbf{u}_{\text{Stokes},h}$, for each $n = 0, 1, \dots$ perform the following steps:

- 1) Find $\tilde{\mathbf{u}}_h = \tilde{\tilde{\mathbf{u}}}_h + \mathcal{R}_h \mathbf{u}_{\text{in}}$, with $\tilde{\tilde{\mathbf{u}}}_h \in V_{0,h}$ such that, $\forall \mathbf{v}_h \in V_{0,h}$,

$$\begin{aligned} & \left(\frac{\tilde{\tilde{\mathbf{u}}}_h}{\Delta t} + (\mathbf{u}_h^n \cdot \nabla) \tilde{\tilde{\mathbf{u}}}_h, \mathbf{v}_h \right) + \frac{2}{\text{Re}} (\varepsilon(\tilde{\tilde{\mathbf{u}}}_h), \varepsilon(\mathbf{v}_h)) \\ &= \left(\frac{\mathbf{u}_h^n}{\Delta t}, \mathbf{v}_h \right) - (\nabla p_h^n, \mathbf{v}_h) + \int_{\Gamma_{\text{out}}} h_{\text{out}} \mathbf{n} \cdot \mathbf{v}_h \, ds \\ & \quad - ((\mathbf{u}_h^n \cdot \nabla) \mathcal{R}_h \mathbf{u}_{\text{in}}, \mathbf{v}_h) - \frac{2}{\text{Re}} (\varepsilon(\mathcal{R}_h \mathbf{u}_{\text{in}}), \varepsilon(\mathbf{v}_h)) \end{aligned}$$

- 2) Find $\delta p_h \in Q_h$ such that, $\forall q_h \in Q_h$,

$$(\nabla \delta p_h, \nabla q_h) = \left(-\frac{1}{\Delta t} \text{div } \tilde{\mathbf{u}}_h, q_h \right).$$

- 3) Project velocity: find $\mathbf{u}_h^{n+1}$ such that

$$(\mathbf{u}_h^{n+1}, \mathbf{v}_h) = (\tilde{\mathbf{u}}_h - \Delta t \nabla \delta p_h, \mathbf{v}_h) \quad \forall \mathbf{v}_h \in V_{0,h}.$$

- 4) Update pressure:

$$p_h^{n+1} = p_h^n + \delta p_h.$$

We point out that step 3 is indeed a projection step, in the L^2 scalar product, of the corrected velocity onto the space V_h . Indeed, the (incremental) pressure gradient $\nabla \delta p_h$

¹Notice that $\frac{1}{\text{Re}} (\nabla \mathbf{u} + \nabla \mathbf{u}^T) = \frac{2}{\text{Re}} \varepsilon(\mathbf{u})$.

does not belong to V_h , thus the quantity $\tilde{\mathbf{u}}_h - \Delta t \nabla \delta p_h$ is not in V_h either, and needs to be projected.

In the implementation, we keep the velocity and pressure spaces separate, since they are used to define different problems, and so we do also with the boundary conditions list. An exception to this separation is the initialization Stokes problem, in which we assemble the monolithic system to have a quick implementation. The forms associated to steps 1, 2, and 3 are implemented in the functions `predictor_forms`, `pressure_forms`, and `projection_forms`, respectively. For each of those steps we create a `LinearVariationalProblem` and a `LinearVariationalSolver`, that is then used in the time loop. Notice that in the definition of the problems and solvers, the implementation choices ensure that the data of each problem is automatically updated at each time step, as long as `u_old` and `p_old` are properly updated.

Solving the problem, we obtain a flow whose final state at $t = 50$ is displayed in Figure 1. The low value of Re allows the flow to be split almost symmetrically by the obstacle in the early part of the time evolution. After some time, however, this symmetry is broken in the wake, and vortex shedding occurs.

We can notice some (small) oscillations of the pressure field near the outlet: this is typical of the differential Chorin-Temam scheme, which introduces a splitting error that is known to determine a boundary layer in the pressure field.

- 2 The forces at each time can be computed as follows:

```
drag_vec[ii] = assemble(
    inner( ph_old*normal - 2.0/Re*dot(sym(grad(uh_old)),
        normal), Constant((1.,0.)) ) * ds(5) )
)
lift_vec[ii]= assemble(
    inner( ph_old*normal - 2.0/Re*dot(sym(grad(uh_old)),
        normal), Constant((0.,1.)) ) * ds(5) )
)
```

resulting in the time-dependent plot reported in Figure 2.

The drag is the force exerted on the object in the direction of the flow, thus it is positive (directed as $+\mathbf{i}$) and it is much larger than the lift force, which acts in the transverse direction. The low value of the lift force is due to the symmetry of the obstacle and the domain, for which very small average transversal force is expected (at least for small Re). The vortex shedding developing in the wake can be seen in the physical oscillations of the lift force.

- 3 The choice of different boundary conditions *may* affect steps 1 and 2 of the projection scheme. Since the predictor step inherits the boundary conditions directly from the main

problem, its new formulation reads

$$\left\{ \begin{array}{ll} \frac{\tilde{\mathbf{u}} - \mathbf{u}^n}{\Delta t} + (\mathbf{u}^n \cdot \nabla) \tilde{\mathbf{u}} - \frac{2}{\text{Re}} \nabla \cdot \varepsilon(\tilde{\mathbf{u}}) = -\nabla p^n, & \text{in } \Omega, \\ \tilde{\mathbf{u}} = (1 - y^2) \mathbf{i}, & \text{on } \Gamma_{\text{in}}, \\ \left(\frac{2}{\text{Re}} \varepsilon(\tilde{\mathbf{u}}) \right) \mathbf{n} = h_{\text{out}} \mathbf{n}, & \text{on } \Gamma_{\text{out}}, \\ \tilde{\mathbf{u}} = \mathbf{0}, & \text{on } \Gamma_{\text{obs}}, \\ \tilde{\mathbf{u}} \cdot \mathbf{n} = 0, \left(\frac{2}{\text{Re}} \varepsilon(\tilde{\mathbf{u}}) \mathbf{n} \right) \cdot \mathbf{t} = 0, & \text{on } \Gamma_{\text{w}}. \end{array} \right.$$

Regarding the boundary conditions for the pressure step, we know that they are derived from the behavior of the normal component of the velocity $\mathbf{u} \cdot \mathbf{n}$. Since it is equal to 0 on Γ_{w} , exactly as in the no-slip case, step 2 is actually unchanged.

In the implementation, we need to change the initialization and the boundary conditions `bcsU` of step 1, and due to this change, we need to re-define the `LinearVariationalProblem` and `LinearVariationalSolver` of step 1. Then, the time loop is unchanged.

With these new boundary conditions, we obtain the flow displayed in Figure 1-right. We can notice how the flow has a different behavior, particularly in the wake and near the lateral walls – as well as lower velocities, since the flowrate is distributed on a flatter profile. Since no wall friction is present, the overall pressure jump between input and output is significantly lower than in the no-slip case. An additional effect is less stable symmetry of the flow, that shows vortex shedding earlier than in the no-slip case (in the simulated timespan $t \in [0, 50]$).

In terms of the forces, we notice how the drag force is lower than in the no-slip case, since there is no “confinement effect” by the walls. The lift force, instead, is significantly affected by the vortex shedding oscillations.

Final remark on symmetry

The symmetry of the wake, particularly in the no-slip case, is much more stable than in real life: although the general comments are true, imperfections in actual physical settings may lead to a slightly different behavior in terms of broken symmetry. Indeed, in Figure 2-left, we can notice that symmetry is slightly broken, being $F_{\text{L}} > 0$ for long times.

In practice, to avoid this artificial feature due to domain being “too symmetric”, meshes are typically constructed with a slight unalignment. Figure 3 reports the results for the same settings of the problem, with no-slip boundary conditions on the walls and obstacle, except for the position of the obstacle, which is slightly displaced upwards (by about $0.5h$): we can see that this slightly broken symmetry allows the wake to develop earlier in time.

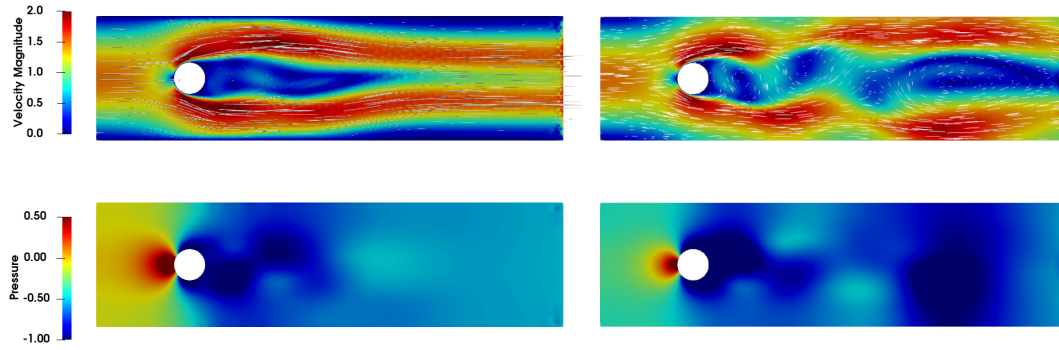


Figure 1: Velocity (top) and pressure (bottom) fields with no-slip (left) and free-slip (right) walls, at $t = 50$.

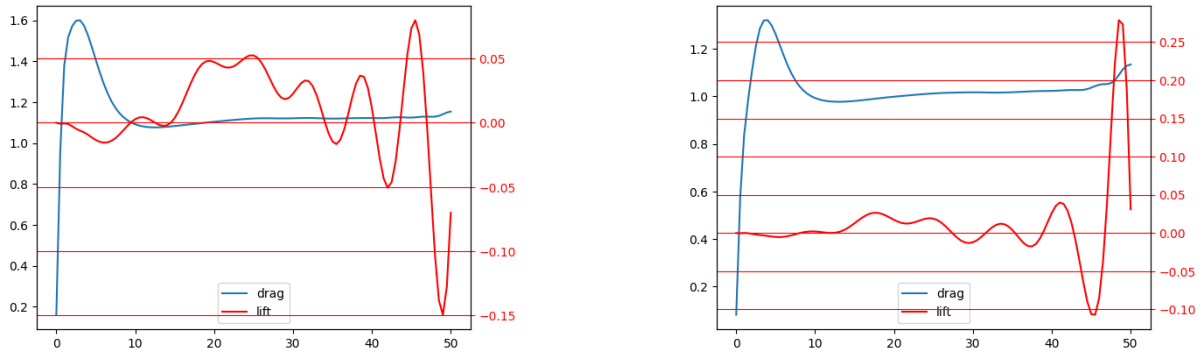


Figure 2: Time evolution of drag and lift forces on the obstacle for no-slip (left) and free-slip (right) walls. ATTENTION: both scales in the left and right plots differ!

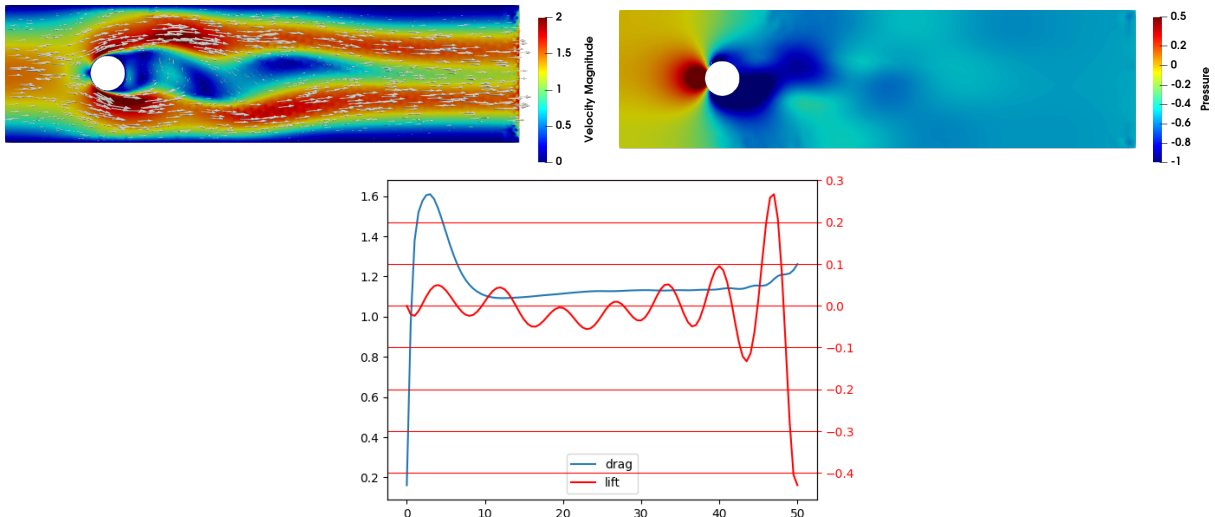


Figure 3: Breaking symmetry: velocity (top left), pressure (top right) and forces on the obstacle (bottom) for no-slip conditions in a slightly unaligned mesh, at $t = 50$.